



Contents lists available at ScienceDirect

Theoretical Computer Science

journal homepage: www.elsevier.com/locate/tcs

The reliability of (n, k) -star network in terms of non-inclusive fault pattern [☆]

Qigong Chen ^{a,b}, Jiafei Liu ^{a,b,*}, Chia-Wei Lee ^c, Jingli Wu ^{a,b}, Gaoshi Li ^{a,b}^a Key Lab of Education Blockchain and Intelligent Technology, Ministry of Education, Guangxi Normal University, Guilin, 541004, China^b Guangxi Key Lab of Multi-Source Information Mining and Security, Guangxi Normal University, Guilin, 541004, China^c Department of Computer Science, University of Taipei, Taipei 100234, Taiwan

ARTICLE INFO

Keywords:

Multiprocessor systems
 (n, k) -star network
Non-inclusive diagnosability
Non-inclusive 1-extra diagnosability
MM* model

ABSTRACT

Reliability assessment is of significant importance in the design maintenance and improvement of multiprocessor systems which take interconnection networks as underlying topologies. System-level diagnosis is a primary strategy to identify faulty processors by analyzing the syndrome of testing in a system. The newly proposed non-inclusive diagnosability greatly enhances the ability in evaluating the reliability of interconnection networks when comparing to the classical diagnosability. In this article, we first study the non-inclusive diagnosability and the non-inclusive 1-extra diagnosability of (n, k) -star network under the MM* model, respectively. Then we design a fast and adaptive non-inclusive fault identification algorithm to identify faulty nodes and fault-free nodes in the (n, k) -star network. Furthermore, we implement simulation experiments in terms of TNR, TPR, ACCR, FNR, FPR and F1. Simulation results demonstrate that our proposed method achieves excellent performance in fault detection and network reliability.

1. Introduction

In multiprocessor systems, several processors are connected by the interconnecting architecture that supports inter-processor data exchange [7,22]. The scale of some supercomputers such as Sunway TaihuLight supercomputer [12] and Tianhe-2 supercomputer [16] has been amplifying dramatically recently, so failures of network components are inevitable with the size of network growing rapidly. To maintain the security and availability of network with multiprocessor systems, a common and effective solution has becomes increasingly important for analyzing network reliability. The fault diagnosis has been regarded as a fundamental technique to assess fault tolerance and reliability of the system against faulty processors.

The field of system-level diagnosis has evolved from the pioneering work of Preparata, Metze, and Chien [17], who established the first diagnostic model, namely, the PMC model in 1967. The PMC model stipulates that each processor can test its neighbors, and the test results are either 0 or 1. The MM model, proposed by Maeng and Malek [15] in 1981, which stipulates that the processor in a system sends the same task to two adjacent processors and compares their feedback results. Latter, Sengupta and Dahbura [18] extended MM model to MM* model, and the MM* model requires that, each processor is capable of testing any pair of processors that are adjacent to it. All possible test results produced under the MM* model are presented in the Table 1. Even to this day, the issue of automatic fault identification has inspired the development of system-level fault diagnosis theory and various diagnosis models [2,3].

[☆] This article belongs to Section A: Algorithms, automata, complexity and games, Edited by Paul Spirakis.^{*} Corresponding author at: Key Lab of Education Blockchain and Intelligent Technology, Ministry of Education, Guangxi Normal University, Guilin, 541004, China.
E-mail address: liujiafei@gxnu.edu.cn (J. Liu).<https://doi.org/10.1016/j.tcs.2025.115189>

Received 14 October 2024; Received in revised form 6 February 2025; Accepted 15 March 2025

0304-3975/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Table 1

Test results of the MM* model.

Comparator	Two tested processors	Result
Faulty-free	Both are faulty-free	0
Faulty-free	At least one is faulty	1
Faulty	Any case	0 or 1

Due to the introduction of graph modeling for system-level diagnosis, numerous studies of diagnosability under the MM* model have been extensively explored by many scholars.

However, traditional diagnosability is generally confined to the minimum node degree of a system. Therefore, Zhang et al. [29] proposed a g -extra condition diagnosability, and applied it to the hypercube network. Cheng et al. [4] explored the g -extra diagnosability of enhanced hypercubes. Wang [20] characterized the g -extra diagnosability of Möbius cubes. However, these diagnosabilities require that two faulty sets are included, i.e., $T_1 \subset T_2$ or $T_2 \subset T_1$. In actual network environments, the probability of $T_1 \subset T_2$ or $T_2 \subset T_1$ occurring in a multiprocessor system is very low. To determine the precise value of diagnosability, we generally construct a pair of smallest indistinguishable sets in the traditional diagnostic schemes. So, taking the (n, k) -star network with $\frac{n!}{(n-k)!}$ nodes as an example, assume that T_1 and T_2 are two faulty sets of the (n, k) -star network with $|T_1| = n$ and $|T_2| = n + 1$. Thus, the probability of

the event $T_1 \subset T_2$ occurring is $\frac{\binom{n!}{(n-k)!}(n+1)}{\binom{n!}{(n-k)!}\binom{n!}{n}}$, whose limit converges to 0 when n approaches infinity. To address this drawback, Ding

et al. [5], proposed the non-inclusive diagnostic scheme, which requires all faulty sets to be non-inclusive, and they characterized the non-inclusive diagnosability of hypercubes under the MM* model. Subsequently, Ma et al. [14] investigated the non-inclusive diagnosability of hypercubes under the PMC model. Wei et al. [19] reported the non-inclusive diagnosability of triangle-free connected graphs, and apply to hypercubes, hierarchical cubes, star graphs, bubble-sort graphs, pancake graphs, and so on. Zhuo et al. [25] studied the non-inclusive diagnosability of alternating group graphs with triangles under PMC model and MM* model. Ding and Xu [6] established the relationship between the diagnosability and non-inclusive diagnosability of triangle-free connected graphs under PMC model. Recently, Yuan et al. [23] developed the non-inclusive g -good-neighbor diagnosability of networks, and applied this scheme to hypercubes, (n, k) -star network as well as (n, k) -bubble-sort graphs. Zhang et al. [24] explored the non-inclusive g -good-neighbor diagnosability of hypercubes. Furthermore, Zheng et al. [30] proved the non-inclusive g -extra diagnosability of (n, k) -star network, DCell, BCDC and BCube under the PMC model. The non-inclusive diagnosability, as a variant of diagnosability, provides a more accurate metric for the reliability of networks compared to existing diagnostic schemes.

The (n, k) -star network $S_{n,k}$, proposed by Chiang and Chen [1], has been known as a famous topology structure of interconnection networks, which is a class of Cayley coset graphs. Later, Feng [8] introduced the (n, k) -star network to address scaling issues of the star graph. Huang et al. [13] employed two algorithms for constructing independent spanning trees in (n, k) -star graphs. Kumar et al. [9] reported the oriented diameter of (n, k) -star graphs, which quantifies an upper limit on the delay in communication over interconnection networks. Lv et al. [11] studied the reliability of (n, k) -star network based on g -extra conditional fault pattern.

Up to day, the research related to non-inclusive diagnostic scheme on (n, k) -star graphs involved very little. In this paper, we explore the non-inclusive diagnosability and non-inclusive g -extra diagnosability of (n, k) -star network under the MM* model. The main contributions of this paper are summarized as follows:

- we theoretically characterize the non-inclusive diagnosability and non-inclusive g -extra diagnosability of (n, k) -star network under the MM* model, respectively;
- based on the characteristics of (n, k) -star graphs and MM* model, we design an adaptive non-inclusive fault diagnosis algorithm for (n, k) -star network under the MM* model;
- we conduct on a series of simulation experiments in terms of the proposed algorithm, and verify experimental results are closely consistent with the theoretical values.

The remainder of this work proceeds as follows. In the next section, we present some necessary terminologies and notions; Section 3 explores the non-inclusive diagnosability and non-inclusive g -extra diagnosability of (n, k) -star network under the MM* model; The adaptive non-inclusive faulty identification algorithm in terms of non-inclusive 1-extra diagnostic strategy is proposed in Section 4. Section 5 implements simulation experiments and verify the correctness of the algorithm. Section 6 ends with conclusions and provides a future outlook.

2. Preliminaries

2.1. Terminologies and notions

In this section, we introduce some terminologies and facts that are needed throughout this paper. An interconnection network is usually modeled by a connected graph G , where nodes represent processors, and edges represent communication links between processors. For a simple and finite graph $G(V, E)$, the notation V and E denote the node set and edge set, respectively. Denote $\langle n \rangle$ by $\{1, 2, \dots, n\}$. For any $v \in V$, we denote $N_G(v) = \{w \in V \mid (v, w) \in E\}$ as the neighborhood of v in G . Moreover, for a simple graph G , the degree of a node v is represented by $d(v) = |N_G(v)|$. The graph G is k -regular graph if and only if $d(v) = k$ for any $v \in V$. For

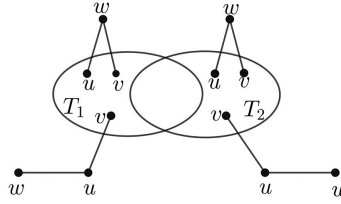


Fig. 1. Illustration of a distinguishable pair (T_1, T_2) under the MM^* model.

a subset S of $V(G)$, $\bar{S}$ represents the complement set of S in G , i.e., $\bar{S} = V(G) \setminus S$. We use $N_G(S)$ to denote the open neighborhood of S , i.e., $N_G(S) = \{w \in V(G) \setminus S \mid v \in S, (v, w) \in E\}$, and $N[S] = N(S) \cup S$ denotes the closed neighborhood. When G is clear from the context, we use $N(v)$ to replace $N_G(v)$, $N(S)$ to replace $N_G(S)$. A subgraph induced by X in G is denoted by $G[X]$. For any two nodes $u, v \in V$, we denote $cn(u, v)$ by the number of common neighbors they share, i.e., $cn(u, v) = |N(u) \cap N(v)|$. Given two node sets $T_1, T_2 \subset V$, the *symmetric difference* T_1 and T_2 is defined as $T_1 \Delta T_2 = (T_1 \setminus T_2) \cup (T_2 \setminus T_1)$. The graph G is a complete graph, denoted by K_n , if and only if there exists an edge between any two distinct nodes u and v in G . A circle refers to a closed path that starts from a node, travels along an edge, and ultimately returns to that starting node, with no nodes being repeated (except for the starting node) in the path. A cycle with h edges $(x_1, x_2, \dots, x_h, x_1)$ is called h -cycle, denoted by C_h . The connectivity of a system G , denoted by $\kappa(G)$, is the minimum size of a node set F such that $G \setminus F$ is disconnected or has only one node. For further information on graph and network, please refer to Ref. [21].

Definition 1. [28] Let T be a faulty set in a system G . If $G \setminus T$ is disconnected and each component include at least $g + 1$ nodes in the residual network, then T is called a g -extra node cut of G . The g -extra connectivity of a system G refers to the minimum size of a node set T , denoted by $\kappa_g(G)$.

Generally, the collection of all faulty nodes is called a faulty set T . Under the MM^* model, the node z sends a same task to two neighbors x and y , and compares their feedback results, denoted by $\sigma((x, y), z)$. Generally, we consider the comparator z to be fault-free. If x and y are fault-free, then $\sigma((x, y), z) = 0$; if at least one of x and y is faulty, then $\sigma((x, y), z) = 1$. For other cases, please refer to Table 1. For a given syndrome σ , if a faulty set T can produce the syndrome σ , we call T is *consistent* with σ . Let $\sigma(T) = \{\sigma \mid T \text{ is consistent with } \sigma\}$. Two different faulty sets T_1 and T_2 are said to be *indistinguishable* if they are consistent with a same syndrome σ , i.e., $\sigma(T_1) \cap \sigma(T_2) \neq \emptyset$; otherwise, T_1 and T_2 are said to be *distinguishable*.

Definition 2. [18] Let $G = (V(G), E(G))$ be a multiprocessor system. For any two faulty sets $T_1, T_2 \subset V(G)$ with $T_1 \neq T_2$, (T_1, T_2) is a distinguishable pair under the MM^* model if and only if at least one of the following conditions is satisfied (see Fig. 1):

- (1) there exists a node $w \in V(G) \setminus (T_1 \cup T_2)$, and $u, v \in T_1 \setminus T_2$ such that $(u, w) \in E(G)$ and $(v, w) \in E(G)$;
- (2) there exists a node $w \in V(G) \setminus (T_2 \cup T_1)$, and $u, v \in T_2 \setminus T_1$ such that $(u, w) \in E(G)$ and $(v, w) \in E(G)$;
- (3) there exist nodes $v, w \in V(G) \setminus (T_1 \cup T_2)$, and $u \in T_1 \Delta T_2$ such that $(u, w) \in E(G)$ and $(v, w) \in E(G)$.

Notably, the non-inclusive diagnosability is a significant indicator to assess the resilience of multiprocessor systems. Two distinct sets T_1 and T_2 are non-inclusive if and only if $T_1 \not\subseteq T_2$ and $T_2 \not\subseteq T_1$. Now, we recall the definition of non-inclusive diagnosability.

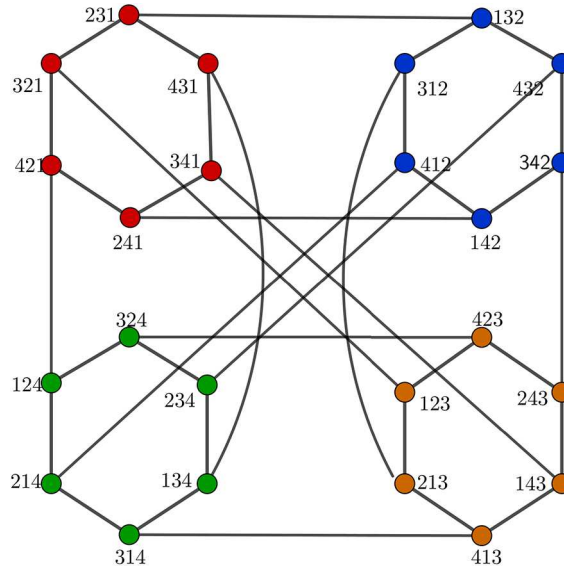
Definition 3. [5] In a multiprocessor system G , for any two non-inclusive faulty sets T_1 and T_2 such that $|T_1| \leq t$, $|T_2| \leq t$, T_1 and T_2 are distinguishable, then G is non-inclusive t -diagnosable. The non-inclusive diagnosability of G , denoted by $t_N(G)$, is the maximum value of t such that G is non-inclusive t -diagnosable.

Definition 4. [30] In a multiprocessor system G , for any two non-inclusive g -extra faulty sets T_1 and T_2 such that $|T_1| \leq t$, $|T_2| \leq t$, T_1 and T_2 are distinguishable, then G is non-inclusive g -extra t -diagnosability. The non-inclusive g -extra diagnosability of G , denoted by $t_N^g(G)$, is the maximum value of t such that G is non-inclusive g -extra t -diagnosable.

2.2. Topological structure of (n, k) -star network $S_{n,k}$

The (n, k) -star network $S_{n,k}$, is a desirable candidate of multiprocessor or multi-computer systems. Compared to the hypercube, $S_{n,k}$ performs many desirable properties, such as node and edge symmetry, high connectivity, small diameter, strong recursive property, and so on.

Definition 5. [1,10,26] The (n, k) -star network $S_{n,k}$, is specified by two integers n and k , where $1 \leq k \leq n - 1$. We denote $V(S_{n,k}) = \{x_1 x_2 \dots x_k \mid x_i \in \langle n \rangle, \text{ and } x_i \neq x_j \text{ for } i \neq j\}$. Let u, v be two nodes in $V(S_{n,k})$ and $u = x_1 x_2 \dots x_i \dots x_k$, then there exists a link between two nodes u and v if one of the following holds (see Fig. 2):

Fig. 2. Illustration of $S_{4,3}$.

- (1) $v = x_i x_2 \cdots x_1 \cdots x_k$, i.e., exchange x_1 with x_i , for $i \in \{2, 3, \dots, k\}$.
- (2) $v = x_p x_2 x_3 \cdots x_k$, i.e., replace x_p with x_1 , where $p \in \{x_{k+1}, x_{k+2}, \dots, x_n\}$.

Furthermore, $S_{n,k}$ is $(n-1)$ -regular, and it has $\frac{n!}{(n-k)!}$ nodes. For any node u and $(u, v) \in E(S_{n,k})$, if the node v satisfies Type (1) in Definition 5, then v is an i -neighbor of u and (u, v) is called an i -edge. If the node v satisfies Type (2) in Definition 5, then v is called an 1-neighbor of u and (u, v) is an 1-edge.

Lemma 2.1. [10] For any two nodes u and v in $S_{n,k}$, we have

$$cn(u, v) = \begin{cases} n-k-1 & \text{if } (u, v) \in E(S_{n,k}) \text{ is one 1-edge;} \\ 1 & \text{if } (u, v) \notin E(S_{n,k}), N_{S_{n,k}}(u) \cap N_{S_{n,k}}(v) \neq \emptyset; \\ 0 & \text{otherwise.} \end{cases}$$

According to Definition 5 and Lemma 2.1, the node u and its all 1-neighbors form a complete graph K_{n-k+1} , for any $u \in V(S_{n,k})$.

Lemma 2.2. [27] Let S be any node subset in $V(S_{n,k})$,

- (1) if $|S| = 2$, then $|N_{S_{n,k}}(S)| \geq n+k-3$;
- (2) if $|S| = 3$, then $|N_{S_{n,k}}(S)| \geq n+2k-5$.

3. Main results

In this section, we establish the non-inclusive diagnosability and non-inclusive 1-extra diagnosability of (n, k) -star networks under the MM* model by a theoretical derivation.

3.1. The non-inclusive diagnosability of $S_{n,k}$ under the MM* model

Firstly, we explore the non-inclusive diagnosability of $S_{n,k}$.

Lemma 3.1. For $n \geq 6, k \geq 3, 2n-k \geq 4$, the upper bound of non-inclusive diagnosability of $S_{n,k}$ under the MM* model is $t_N(S_{n,k}) \leq n+k-3$.

Proof. Let T_1 and T_2 be non-inclusive in $S_{n,k}$ under the MM* model. Take two nodes $u, v \in V(S_{n,k})$ and $(u, v) \in E(S_{n,k})$. Assume that $T_1 = N(\{u, v\}) \cup \{u\}$, $T_2 = N(\{u, v\}) \cup \{v\}$. Then $|T_1| = |T_2| = |N(\{u, v\})| + 1 = (n+k-3) + 1 = n+k-2$. Notice that there is no edge between $T_1 \triangle T_2$ and $T_1 \cup T_2$. By Definition 2, T_1 and T_2 are indistinguishable under the MM* model. So, we have $t_N(S_{n,k}) \leq n+k-3$.

Lemma 3.2. For $n \geq 6, k \geq 3, 4n-k \geq 4$, the lower bound of non-inclusive diagnosability of $S_{n,k}$ under the MM* model is $t_N(S_{n,k}) \geq n+k-3$.

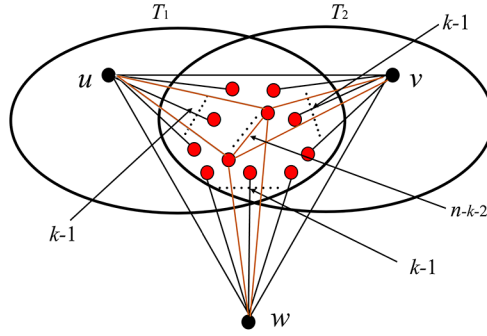


Fig. 3. Illustration of $|N(T_1 \triangle T_2) \cap \overline{T_1 \cup T_2}| = 1$.

Proof. Now we determine the lower bound of $t_N(S_{n,k})$ by a contradiction method. Suppose that T_1 and T_2 are non-inclusive and indistinguishable under the MM* model. To this aim, we will prove $|T_1| \geq n+k-2$ or $|T_2| \geq n+k-2$. Without loss of generality, let $|T_1| \geq |T_2|$. Since T_1 and T_2 are indistinguishable, take a node $u \in T_1 \triangle T_2$, such that $|N(u) \cap \overline{T_1 \cup T_2}| \geq |N(v) \cap \overline{T_1 \cup T_2}|$ for any $v \in T_1 \triangle T_2$. According to the distribution of $N(u)$, we divide into four cases as below:

Case 1: $|N(u) \cap \overline{T_1 \cup T_2}| = 0$.

Since T_1 and T_2 are a pair of non-inclusive sets, we have $|T_1 \triangle T_2| \geq 2$. Then, according to the size of $T_1 \triangle T_2$, the following possibilities can be discussed.

Subcase 1.1: $|T_1 \triangle T_2| = 2$.

Take $T_1 \triangle T_2 = \{u, v\}$, so $N(\{u, v\}) \subset T_1 \cap T_2$. Since T_1 and T_2 are non-inclusive, $|T_1 \setminus T_2| = 1$. By Lemma 2.2, then we have

$$|T_1| \geq |T_1 \cap T_2| + 1 \geq |N(\{u, v\})| + 1 \geq n+k-3+1 = n+k-2.$$

Subcase 1.2: $|T_1 \triangle T_2| \geq 3$.

In this case, let $u, v, w \in T_1 \triangle T_2$. Now we discuss $T_1 \cup T_2 = V(S_{n,k})$ and $T_1 \cup T_2 \neq V(S_{n,k})$.

Subcase 1.2.1: $T_1 \cup T_2 = V(S_{n,k})$.

Because of $|V(S_{n,k})| = \frac{n!}{(n-k)!}$, we have

$$|T_1| \geq \left\lceil \frac{|T_1 \cup T_2| + |T_1 \cap T_2|}{2} \right\rceil \geq \left\lceil \frac{|V(S_{n,k})|}{2} \right\rceil \geq \left\lceil \frac{n!}{2(n-k)!} \right\rceil \geq n+k-2,$$

which holds for $n \geq 3, n-k \geq 1$.

Subcase 1.2.2: $T_1 \cup T_2 \neq V(S_{n,k})$.

If $T_1 \cap T_2 = \emptyset$ and $|N(u) \cap \overline{T_1 \cup T_2}| = 0$, then $N(T_1 \triangle T_2) = \emptyset$. One can observe that, $T_1 \cup T_2$ is not connected to $\overline{T_1 \cup T_2}$, which contradicts that $S_{n,k}$ is connected. Hence, we have $|T_1 \cap T_2| \geq 1$. By Lemma 2.2, we have $|T_1 \cup T_2| \geq |N(\{u, v, w\})| + |\{u, v, w\}| \geq n+2k-5+3 = n+2k-2$. To this regard, if $|T_1 \cap T_2| \geq n-1$, then

$$|T_1| \geq \left\lceil \frac{|T_1 \cup T_2| + |T_1 \cap T_2|}{2} \right\rceil \geq \left\lceil \frac{n+2k-2+n-1}{2} \right\rceil \geq n+k-2.$$

Next we consider the case of $1 \leq |T_1 \cap T_2| \leq n-2$. By means of Lemma 2.2, we obtain $|N(\{u, v, w\})| \geq n+2k-5$. Since $n+2k-5-(n-2) \geq 3(k \geq 3)$, there are at least three nodes in $(T_1 \triangle T_2) \setminus \{u, v, w\}$. So, we have $|T_1 \cup T_2| \geq 6$. Then, $|T_1 \cup T_2| \geq |N[K_6]| = n-k+1+6(k-1) = n+5k-5$. Hence,

$$|T_1| \geq \left\lceil \frac{|T_1 \cup T_2| + |T_1 \cap T_2|}{2} \right\rceil \geq \left\lceil \frac{n+5k-5+1}{2} \right\rceil \geq \left\lceil \frac{n+5k-4}{2} \right\rceil \geq n+k-2,$$

which holds for $n \leq 3k$.

Case 2: $|N(u) \cap \overline{T_1 \cup T_2}| = 1$.

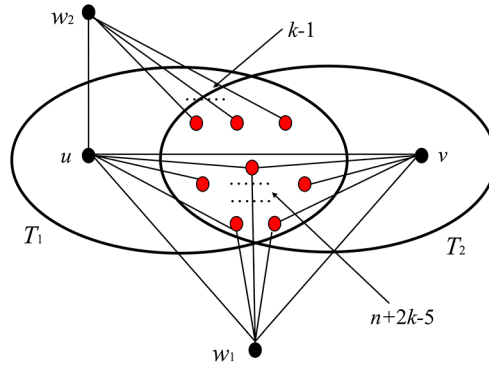
In this case, we consider all possibilities of $N(T_1 \triangle T_2) \cap \overline{T_1 \cup T_2}$ as below.

Subcase 2.1: $|N(T_1 \triangle T_2) \cap \overline{T_1 \cup T_2}| = 1$.

Let $N(T_1 \triangle T_2) \cap \overline{T_1 \cup T_2} = \{w\}$, $v \in T_2 \setminus T_1$ (see Fig. 3). Since T_1 and T_2 are indistinguishable, all neighbors of w are in $T_1 \cap T_2$. So, $|T_1 \cap T_2| \geq |N(w)| - 2 \geq (n-1) - 2 = n-3$. Additionally, $|N(u) \cap \overline{T_1 \cup T_2}| = 1$. By Lemma 2.2, we obtain

$$|T_1 \cup T_2| \geq |N(\{u, v, w\})| + |\{u, v\}| \geq n+2k-5+2 = n+2k-3.$$

If $|T_1 \cap T_2| \geq n-1$, then $|T_1| \geq \left\lceil \frac{|T_1 \cap T_2| + |T_1 \cup T_2|}{2} \right\rceil \geq n+k-2$. Otherwise, $n-3 \leq |T_1 \cap T_2| \leq n-2$. By Lemma 2.2, we have $|N(\{u, v, w\})| \geq n+2k-5$. When $k \geq 3$, $|N(\{u, v, w\})| - |T_1 \cap T_2| = n+2k-5-(n-2) \geq 3$. Noting that T_1 and T_2 are indistinguishable under the MM* model, then $N(\{u, v, w\}) \subset T_1 \cup T_2$. Therefore, apart from nodes u and v , there are at least three nodes in $T_1 \triangle T_2$, marked as x, y, z .

Fig. 4. Illustration of $|N(u) \cap \overline{T_1 \cup T_2}| = 2$.

Let $X = \{u, v, w, x, y, z\}$. Accordingly, $|T_1 \cup T_2| \geq |N(X)| + 5 = n - k + 1 - 6 + 6(k - 1) + 5 = n + 5k - 11 + 5 = n + 5k - 6$. Hence, we have

$$\begin{aligned} |T_1| &\geq \left\lceil \frac{|T_1 \cup T_2| + |T_1 \cap T_2|}{2} \right\rceil \\ &\geq \left\lceil \frac{(n + 5k - 6) + (n - 3)}{2} \right\rceil \\ &\geq n + k - 2 \quad (k \geq 2). \end{aligned}$$

Subcase 2.2: $|N(T_1 \triangle T_2) \cap \overline{T_1 \cup T_2}| \geq 2$.

Let $w_1, w_2 \in N(T_1 \triangle T_2) \cap \overline{T_1 \cup T_2}$. Without loss of generality, let (u, w_1) and (v, w_2) be two edges in $E(S_{n,k})$. Since T_1 and T_2 are indistinguishable, $(N(\{w_1, w_2\}) - T_1 \triangle T_2) \subseteq T_1 \cup T_2$. Therefore, $|T_1 \cap T_2| \geq |N_{T_1 \cap T_2}(\{w_1, w_2\})| \geq n - 2 + n - 2 - 1 = 2n - 5$. In terms of $|N(u) \cap \overline{T_1 \cup T_2}| = 1$, one can observe that the neighbors of u and v are in $T_1 \cup T_2$. If both of (u, w_1) and (v, w_2) are 1-edges, then we obtain that

$$\begin{aligned} |T_1 \cup T_2| &\geq |N(\{u, v, w_1, w_2\})| + |\{u, v\}| \\ &\geq 2(n - k - 1 + 2(k - 1) - 1) + 2 \\ &= 2n + 2k - 6. \end{aligned}$$

Thus, we have $|T_1| \geq \left\lceil \frac{|T_1 \cup T_2| + |T_1 \cap T_2|}{2} \right\rceil \geq \left\lceil \frac{(2n + 2k - 6) + (2n - 5)}{2} \right\rceil \geq n + k - 2$, which holds for $n \geq 4$.

Case 3: $|N(u) \cap \overline{T_1 \cup T_2}| = 2$.

Let $N(u) \cap \overline{T_1 \cup T_2} = \{w_1, w_2\}$ (see Fig. 4). Because T_1 and T_2 are indistinguishable, $(N(\{w_1, w_2\}) - T_1 \triangle T_2) \subseteq T_1 \cup T_2$. As a result, $|T_1 \cap T_2| \geq |N_{T_1 \cap T_2}(\{w_1, w_2\})| \geq (n - 3) + (n - 2) = 2n - 5$. By Lemmas 2.1 and 2.2, we have

$$\begin{aligned} |T_1 \cup T_2| &\geq |N(\{u, v, w_1, w_2\})| + |\{u, v\}| \\ &\geq (n + 2k - 5) + (n - 2) + 2 \\ &= 3n + 2k - 5. \end{aligned}$$

Then, $|T_1| \geq \left\lceil \frac{|T_1 \cup T_2| + |T_1 \cap T_2|}{2} \right\rceil \geq \left\lceil \frac{(3n + 2k - 5) + (2n - 5)}{2} \right\rceil > n + k - 2$, which holds for $n \geq 2$.

Case 4: $|N(u) \cap \overline{T_1 \cup T_2}| \geq 3$.

Let $T' \subset (N(u) \cap \overline{T_1 \cup T_2})$, and $|T'| = 3$. Without loss of generality, take $T' = \{w_1, w_2, w_3\}$. By Lemma 2.1, w_1, w_2 and w_3 have at most one common neighbor u , and one of them has an edge with node v . Then, we have

$$|T_1| \geq |N(\{w_1, w_2, w_3\})| + |\{u\}| \geq (2(n - 2) + (n - 3)) + 1 \geq n + k - 2,$$

which holds for $2n - k \geq 4$. Thus, we have $|T_2| \geq n + k - 2$.

Therefore, in the case where T_1 and T_2 are distinguishable and non-inclusive, $t_N(S_{n,k}) \geq n + k - 3$, as required.

Applying Lemmas 3.1 and 3.2, we determine the next result.

Theorem 1. For $n \geq 6, k \geq 3, 2n - k \geq 4$, the non-inclusive diagnosability of $S_{n,k}$ under the MM* model is $t_N(S_{n,k}) = n + k - 3$.

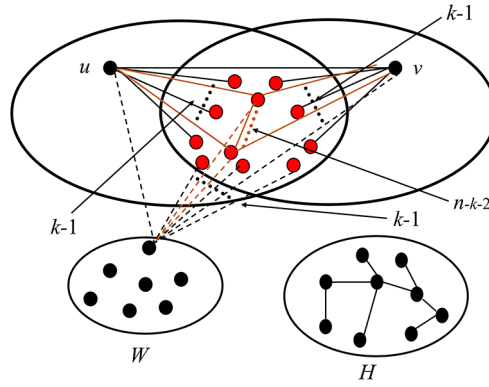
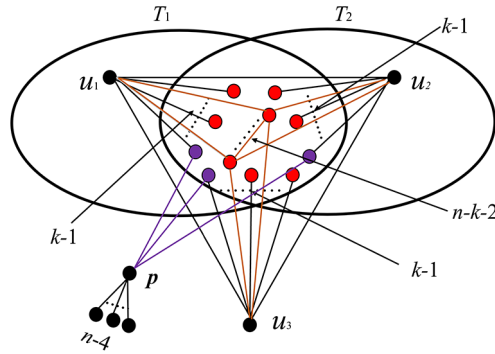
Fig. 5. Illustration of W and H in Lemma 3.4.

Fig. 6. Illustration of Lemma 3.3.

3.2. The non-inclusive 1-extra diagnosability of $S_{n,k}$ under the MM^* model

In this section, we characterize the non-inclusive 1-extra diagnosability of $S_{n,k}$ under the MM^* model.

Lemma 3.3. For $n \geq 5, k \geq 2, n - k \geq 3$, the upper bound of non-inclusive 1-extra diagnosability of $S_{n,k}$ under the MM^* model is $t_N^1(S_{n,k}) \leq n + 2k - 5$.

Proof. Let $A = \{u_1, u_2, u_3\}$ be a subset in $V(S_{n,k})$ such that the induced subgraph $S_{n,k}[A]$ is a 3-cycle. Let $T_1 = N_{S_{n,k}}(A) \cup \{u_1\}$, $T_2 = N_{S_{n,k}}(A) \cup \{u_2\}$, $u_3 \in T_1 \cup T_2$ (see Fig. 6). Then $|T_1| = (n - k + 1 - 3 + 3(k - 1)) + 1 = n + 2k - 4$. Similarly, $|T_2| = n + 2k - 4$. Let p be any node in $T_1 \cap T_2$. By Lemma 2.1, each of u_1, u_2 and u_3 has at most a common neighbor with p . So, p has at most three common neighbors in $T_1 \cap T_2$. When $n \geq 5$, p has at least one neighbor in $T_1 \cup T_2$. Thus, the minimum degree of each node in $V(S_{n,k}) \setminus (T_1 \cup T_2 \cup A)$ is at least one. Hence, T_1 and T_2 are two non-inclusive 1-extra node cuts under the MM^* model. By Definition 2, we conclude that T_1 and T_2 are two indistinguishable sets. Therefore $t_N^1(S_{n,k}) \leq n + 2k - 5$.

Lemma 3.4. For $n \geq 5, k \geq 2, n - k \geq 3$, the lower bound of non-inclusive 1-extra diagnosability of (n, k) -star network under the MM^* model is $t_N^1(S_{n,k}) \geq n + 2k - 5$.

Proof. Suppose, to the contrary, that $t_N^1(S_{n,k}) \geq n + 2k - 4$. By Definition 2, there exists an indistinguishable non-inclusive 1-extra faulty pair (T_1, T_2) such that $T_1 \neq T_2$ and $|T_1| \leq n + 2k - 5$ and $|T_2| \leq n + 2k - 5$. Notice that

$$|T_1 \cup T_2| \leq |T_1| + |T_2| \leq 2n + 4k - 10 \leq \frac{n!}{(n - k)!},$$

which holds for $n \geq 3, k \geq 2, n - k \geq 3$. Thus, $T_1 \cup T_2 \neq V(S_{n,k})$. Assuming that W is an isolate node set of $V(S_{n,k}) \setminus (T_1 \cup T_2)$, $H = V(S_{n,k}) \setminus (T_1 \cup T_2 \cup W)$ (see Fig. 5). Now, it suffices to prove that $W \neq \emptyset$. Let $w \in W$, then $(N_{S_{n,k}}(w) \setminus T_1) \subset (T_2 \setminus T_1)$. Because T_1 is a non-inclusive 1-extra faulty node cut, w has at least one neighbor in $V(S_{n,k}) \setminus T_1$. Furthermore, because T_1 and T_2 are indistinguishable, it follows that $|N_{S_{n,k}}(w) \cap (T_1 \setminus T_2)| = 1$. Similarly, we obtain $|N_{S_{n,k}}(w) \cap (T_2 \setminus T_1)| = 1$. So $|N_{T_1 \cap T_2}(w)| = n - 3$. Thus, we have

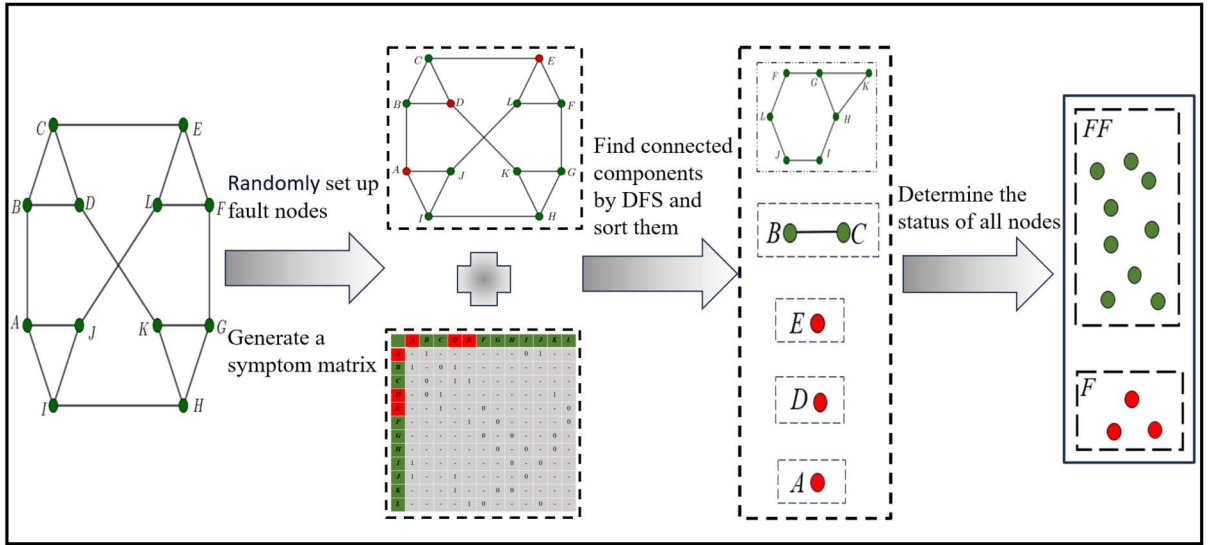


Fig. 7. Illustration of adaptive non-inclusive fault identification algorithm ANFI for (n, k) -star network.

$$\begin{aligned}
 \sum_{w \in W} |N_{T_1 \cap T_2}(w)| &= |W|(n-3) \\
 &\leq \sum_{v \in T_1 \cap T_2} |T_1 \cap T_2|(n-1) \\
 &\leq (|T_1| - 1)n \\
 &\leq (n + 2k - 5 - 1)(n - 1) \\
 &= (n + 2k - 6)(n - 1).
 \end{aligned}$$

As a result, $|W| \leq \frac{(n+2k-6)(n-1)}{n-3}$.

If $V(H) = \emptyset$, then $T_1 \cup T_2 \cup W = V(S_{n,k})$, and we have

$$\begin{aligned}
 \frac{n!}{(n-k)!} &= |T_1 \cup T_2| + |W| \\
 &\leq |T_1| + |T_2| + |W| \\
 &\leq 2(n + 2k - 5) + \frac{(n + 2k - 6)(n - 1)}{n - 3},
 \end{aligned} \tag{1}$$

which is a contradiction for $n \geq 5, k \geq 2, n - k \geq 3$. This implies that $H \neq \emptyset$.

Because T_1 and T_2 are indistinguishable, it follows that $N((T_1 \triangle T_2) \cup W) \subset T_1 \cap T_2$. Since T_1 and T_2 are a pair of non-inclusive sets, then we have $|T_1 \setminus T_2| \geq 1$ and $|T_2 \setminus T_1| \geq 1$. Hence, $|T_1 \cap T_2| \leq n + 2k - 6$. Note that there are at least three nodes in $(T_1 \triangle T_2) \cup W$. Let $S = (T_1 \triangle T_2) \cup W$. By Lemma 2.2, $|T_1 \cap T_2| \geq |N(S)| \geq n + 2k - 5$, which is a contradiction. So, $|S| \leq 2$. Thus, we have $W = \emptyset$.

Because T_1 and T_2 are a pair of non-inclusive 1-extra faulty sets, $|T_1 \setminus T_2| \geq 2$ and $|T_2 \setminus T_1| \geq 2$. Therefore, $|T_1 \cap T_2| \geq |N(K_4)| = n - k + 1 - 4 + 4(k - 1) = n + 3k - 7$. Then, we have

$$|T_1| = |T_1 \cap T_2| + |T_1 \setminus T_2| \geq n + 3k - 7 + 2 = n + 3k - 5 > n + 2k - 5,$$

which is a contradiction. Therefore, $t_N^1(S_{n,k}) \geq n + 2k - 5$, as required.

Applying Lemmas 3.3 and 3.4, we determine the next result.

Theorem 2. For $n \geq 5, k \geq 2, n - k \geq 3$, the non-inclusive 1-extra diagnosability of $S_{n,k}$ under the MM^* model is $t_N^1(S_{n,k}) = n + 2k - 5$.

4. Adaptive non-inclusive fault identification algorithm for (n, k) -star network

In this section, we design an adaptive non-inclusive fault identification algorithm (for short ANFI) to identify all faulty nodes of (n, k) -star network under the MM^* model (see Algorithm 1). The flowchart of this algorithm follows Fig. 7. First, the ANFI algorithm randomly generates faulty nodes for a graph to be detected, and the symptom set of the graph is updated accordingly. Second, using Depth-First Search (DFS) with the symptom set, find all connected components. Third, sort the components, and put the nodes in the

largest component into the fault-free node set. Based on MM* model, detect the status of all other nodes. It mainly consists of three steps: Initialization, Component generation, as well as Faulty node identification.

Initialization (lines 1-6): Construct a list vn_1 to record the visited nodes, a list cp to record a connected component, a list cps to record all connected components, a faulty node list f and a fault-free node list ff to store the diagnosed faulty and fault-free nodes, respectively, along with a list vd_2 to mark whether a node has been diagnosed.

Component generation (lines 7-19): In this process, we search for all connected components of (n, k) -star network by the DFS algorithm when given a syndrome matrix in terms of PMC model. Initially, we randomly choose a node in $S_{n,k}$, and take it into vn_1 and cp . Then, visit one of unvisited neighbors of u , marked as v . If the test result $\sigma(u, v) = 0$, then v is marked as the visited node and put it into the list cp . If $\sigma(u, v) = 1$, then v is marked as the visited node and put it into the list vn_1 . Then continue to the above steps to visited one of unvisited neighbors of v . This process ends until some vertex is accessed, and its all neighbors have already been visited. Thus, we obtain a connected component, and put this component into the list cps . Similar processes are conducted until there are no more nodes can be visited. Finally, sort all connected components of the list cps in descending order according to the size of each component, marked as $cps[1], cps[2], \dots, cps[M]$.

Algorithm 1: Adaptive Non-inclusive Faulty Identification Algorithm ANFI.

Input: an (n, k) -star network $S_{n,k}$, the symptom matrix sm , theoretical value $t_N(S_{n,k})$.
Output: Fault-free nodes set ff , faulty nodes set f

```

1   $vn_1 \leftarrow \emptyset$ ; // All nodes as unvisited.
2   $vn_2 \leftarrow \emptyset$ ;
   // Used to mark whether a node has been diagnosed
3   $cp \leftarrow \emptyset$ ;
4   $cps \leftarrow \emptyset$ ;
5   $ff \leftarrow \emptyset$ ; // Initialize the fault-free nodes set  $ff$ .
6   $f \leftarrow \emptyset$ ; // Initialize the faulty nodes set  $f$ .
7   $vn_1.add(u)$ ; // Mark visited node.
8   $cp.add(u)$ ; // Create the component where the node  $u$  is located.
9  DFS( $node, V, sm, vn_1$ )
10  if  $sm[node][nv[node]] == 0$  and  $nv[node]$  not in  $vn_1$  then
11  |   DFS( $nv[node], V, sm, vn_1$ )
12  while  $x$  in  $nv[u]$  do
13  |   //  $nv[u]$  is the neighbor set of  $u$ 
14  |   if  $sm[u][x] == 0$  and  $x$  not in  $vn_1$  then
15  |   |    $cp.add(x)$ ;
16  |   |   DFS( $x, V, sm, vn_1$ );
17   $cps.append(cp)$ ; // Add the obtained connected component to the set of connected components.
18   $cps.sort()$ ; // Sort the obtained components by quicksort.
19   $ff \leftarrow cps[0]$ ; // The largest component that consists entirely of fault-free nodes.
20   $vd_2 \leftarrow cp[i]; i \leftarrow 1$ ; // Start the connected component set from the second element.
21  while  $len(cps[i]) > t_N(S_{n,k})$  do
22  |    $ff \leftarrow cp[i]$ ; // Considered that all nodes in  $cp[i]$  are fault-free.
23  |    $vn_2 \leftarrow cp[i]; i \leftarrow i + 1$ ;
24  while  $len(f) \leq t_N(S_{n,k})$  do
25  |   while  $u, w \in ff$  do
26  |   |   while  $v \in nv[u]$  do
27  |   |   |   if  $\sigma(u, v) = 0$  and  $\sigma(u, w) = 0$  and  $v$  not in  $vn_2$  then
28  |   |   |   |    $ff \leftarrow v$ ;
29  |   |   |   |    $vn_2 \leftarrow v$ ;
30  |   |   |   else
31  |   |   |   |    $f \leftarrow v$ ;
32  |   |   |   |    $vn_2 \leftarrow v$ ;
33  return  $ff, f$ ;
```

Faulty node identification (lines 20-31): Choose the largest component $cps[1]$ in cps . If the size of $cps[1]$ is more than or equal to the theoretical value $t_N(S_{n,k})$, we take all nodes of $cps[1]$ as fault-free, add them into the list ff . Then, start from nodes u in $cps[1]$ and search for a neighbor w of u in $cps[1]$. Then, find a neighbor v in unknown nodes. Based on MM* model, we obtain the test results for a given node by sending the same task to its two adjacent nodes, and then comparing the results from its two neighbors, if $\sigma((w, v), u) = 0$, the node v is added to the list ff and list vn_2 . If $\sigma((w, v), u) = 1$, the node v is added to the list f and list vn_2 . If the size of list f is equal to $t_N(S_{n,k})$, add other unknown nodes to list ff . Otherwise, we perform the similar process on other components until the size of list f is equal to $t_N(S_{n,k})$.

In the following, we will analyze the time complexity of ANFI to verify the performance of our algorithm.

Lemma 4.1. Let N be the number of nodes in $S_{n,k}$, then the time complexity of ANFI is $O(N^2)$.

Proof. The time complexity of executing the algorithm ANFI is analyzed as below, wherein M is the amount of components and $M \leq N$.

- Steps 1-8: In the initialization stage, it has the time complexity of $O(1)$.
 - Steps 9-11: Depth-First Search traverses the entire graph with the time complexity of $O(N)$.
 - Steps 12-15: Find all connected components with a time complexity of $O(N^2)$.
 - Steps 16-17: Rank all components of the list cps in descending order based on the size of components, with a time complexity of $O(M \log_2 M)$.
 - Steps 18-19: Mark the nodes in the largest component as fault-free nodes and visited nodes with a time complexity of $O(1)$.
 - Steps 20-22: Mark the nodes in the component with more nodes than the diagnosability as fault-free nodes and visited nodes with a time complexity of $O(M)$.
 - Steps 23-31: Identify other faulty nodes and fault-free nodes by the symptom matrix with a time complexity of $O(N^2)$.
- Therefore, the time complexity of ANFI algorithm takes $O(1 + N + N^2 + M \log_2 M + 1 + M + N^2) = O(N^2)$.

5. Experimental simulation

In this section, we will implement simulation experiments and validate the performance of our proposed algorithm ANFI for (n, k) -star network, which is conducted on Python 3.12.4 with the NetworkX package. In the experiments, we introduce six different metrics, including True Negative Rate (TNR), True Positive Rate (TPR), Accuracy (ACCR), False Negative Rate (FNR), False Positive Rate (FPR), as well as F1-score(F1), and their definitions are depicted as below.

- TNR is the rate of correctly identified the fault-free nodes to that of all fault-free nodes,

$$TNR = \frac{TN}{FP + TN}.$$

- TPR is the rate of correctly identified faulty nodes to that of all faulty nodes,

$$TPR = \frac{TP}{FN + TP}.$$

- ACCR is the rate of the number of correctly identified the nodes to that of all nodes,

$$ACCR = \frac{TN + TP}{TN + TP + FN + FP}.$$

- FNR is the rate of the number of incorrectly identified fault-free nodes to that of all fault-free nodes,

$$FNR = \frac{FN}{FP + TN}.$$

- FPR represents the proportion of fault-free nodes misdiagnosed as faulty to all fault-free nodes,

$$FPR = \frac{FP}{FP + TN}.$$

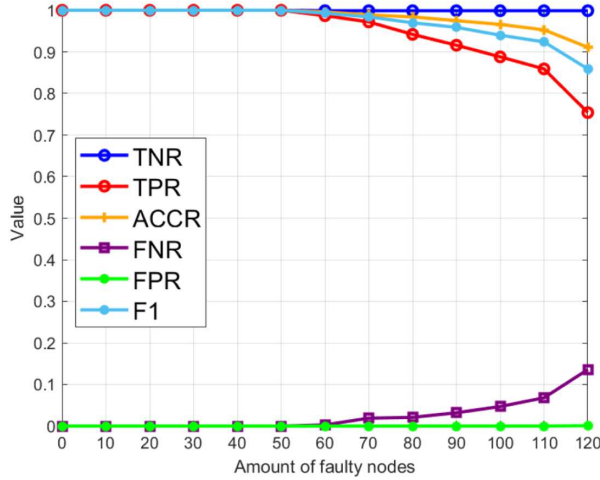
- F1 provides a balanced assessment of the model's performance via integrating Precision with Recall,

$$F1 = 2 * \frac{\frac{TP}{TP+FP} * TPR}{\frac{TP}{TP+FP} + TPR}.$$

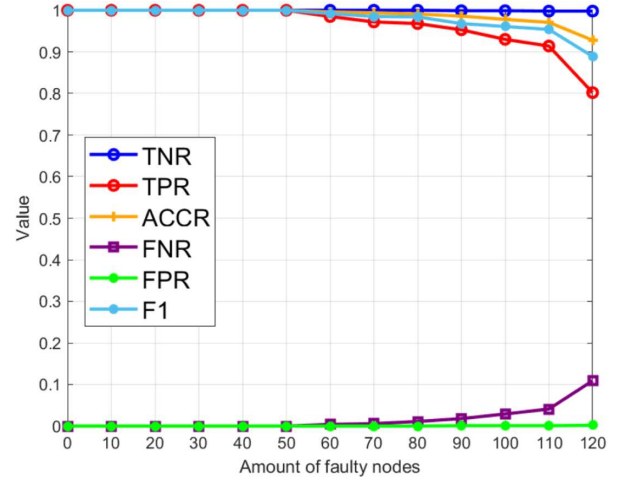
Among them, TP represents the number of correctly identified faulty nodes, TN represents the number of correctly identified fault-free nodes, FP represents the number of incorrectly identified as faulty, and FN represents the number of incorrectly identified as fault-free. In the experiments, we randomly set the number of faulty nodes in terms of TNR, TPR, ACCR, FNR, FPR as well as F1, then perform 5000 times, and calculate their average values.

Figs. 8(a) and (b) demonstrate the experimental results associated with TNR, TPR, ACCR, FNR, FPR as well as F1 of $S_{8,3}$ in terms of $t_N(S_{n,k})$ and $t_N^1(S_{n,k})$, respectively. The corresponding results are presented in Table 2 and Table 3. As observed in Figs. 8(a) and Table 2, with the number of faulty nodes increasing to 60, the values of TNR, TPR, ACCR and F1 remain closed to 1, whereas the values of FNR approach to 0, which indicates that our algorithm can be capable of accurately and efficiently detecting all faulty nodes, thereby ensuring efficient fault diagnosis. In addition, when the number of faulty nodes is no more than 100, TPR is relatively high, which is always above 88.8%. Also, when the number of faulty nodes attains 120, TPR is still higher than 75.4%. Notably, it is noted from Table 2 that TNR is always higher than 99.9% with the amount of faulty nodes increasing from 0 to 120. As the number of faulty nodes is increasingly growing from 60 to 120, ACCR and F1 is not below 85.9% while the value of FNR is less than 13.6%. This suggests that the non-inclusive diagnostic strategy exhibits a high levels of fault recognition and robustness for a system, regardless of the rapid growth of fault nodes.

In Figs. 8(b) and Table 3, it shows a relationship between the number of faulty nodes and distinct metrics under non-inclusive 1-extra diagnostic strategy. As reported in the aforementioned theoretical analysis (follow Section 3), the non-inclusive 1-extra diagnosability is $t_N^1(S_{n,k}) = n + 2k - 5$, especially, when $n = 8$, $k = 3$, we have $t_N^1(S_{n,k}) = 9$. One can observe that, when the number of

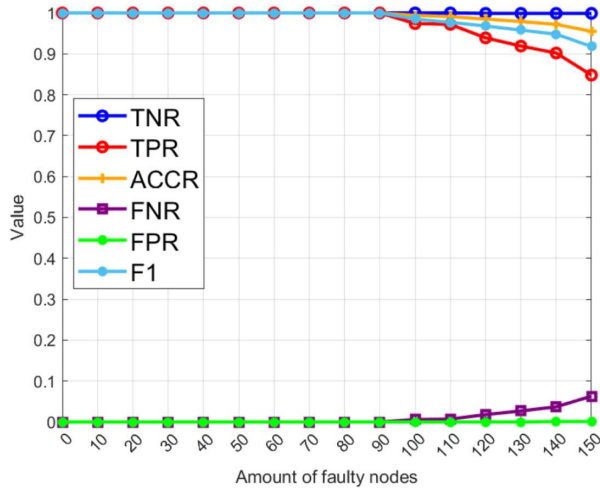


(a) TNR, TPR, ACCR, FNR, FPR and F1 of $S_{8,3}$ with different amount of faulty nodes about non-inclusive diagnosability.

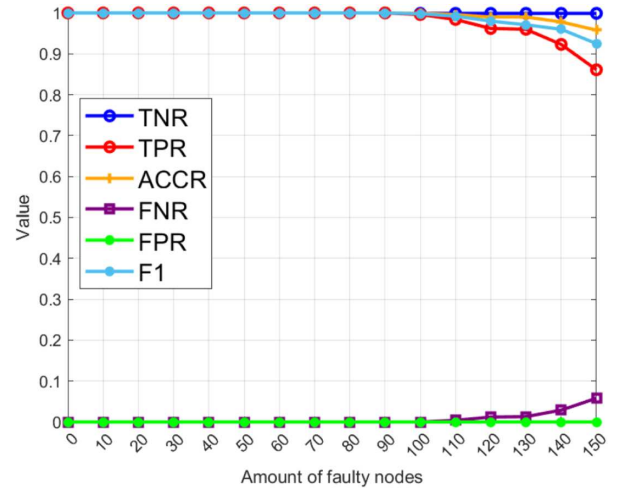


(b) TNR, TPR, ACCR, FNR, FPR and F1 of $S_{8,3}$ with different amount of faulty nodes about non-inclusive 1-extra diagnosability.

Fig. 8. Illustration of TNR, TPR, ACCR, FNR, FPR and F1. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)



(a) TNR, TPR, ACCR, FNR, FPR and F1 of $S_{9,3}$ with different amount of faulty nodes about non-inclusive diagnosability.



(b) TNR, TPR, ACCR, FNR, FPR and F1 of $S_{9,3}$ with different amount of faulty nodes about non-inclusive 1-extra diagnosability.

Fig. 9. Illustration of TNR, TPR, ACCR, FNR, FPR and F1.

Table 2

The non-inclusive diagnosability of $S_{8,3}$, TNR, TPR, ACCR, FNR, FPR, as well as F1 under different faulty nodes.

$ F $	0	20	40	60	80	100	120
TNR	100%	100%	100%	99.9%	99.9%	99.9%	99.9%
TPR	100%	100%	100%	98.7%	94.2%	88.8%	75.4%
ACCR	100%	100%	100%	98.4%	96.6%	95.3%	91.1%
FNR	0.00%	0.00%	0.00%	0.30%	2.10%	4.70%	13.6%
FPR	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.10%
F1	100%	100%	100%	99.3%	97.0%	94.0%	85.9%

faulty nodes grow increasingly (no more than 120), TNR, TPR, ACCR and F1 exhibit a slight declining trend, whose values are all above 80%. While FNR and FPR perform the slightly upward trend below 10.9% despite the escalating amount of faulty nodes.

Figs. 9(a) and (b) showcase the experimental outcomes pertaining to the performance metrics of $S_{9,3}$ under two diagnostic strategies. Compared to $S_{8,3}$, $S_{9,3}$ possesses the larger scale in node amount. As reported in Section 3, the non-inclusive diagnosability and non-inclusive 1-extra diagnosability of $S_{9,3}$ are $t_N(S_{9,3}) = 9$, $t_N^1(S_{9,3}) = 10$, respectively. When the number of faulty nodes is less than

Table 3

The non-inclusive 1-extra diagnosability of $S_{8,3}$, TNR, TPR, ACCR, FNR, FPR, as well as F1 under different faulty nodes.

$ F $	0	20	40	60	80	100	120
TNR	100%	100%	100%	100%	100%	99.9%	99.8%
TPR	100%	100%	100%	98.5%	96.8%	93.0%	80.2%
ACCR	100%	100%	100%	99.6%	99.1%	97.8%	92.8%
FNR	0.00%	0.00%	0.00%	0.40%	1.10%	2.90%	10.9%
FPR	0.00%	0.00%	0.00%	0.00%	0.00%	0.10%	0.20%
F1	100%	100%	100%	99.2%	98.4%	96.1%	88.9%

Table 4

The non-inclusive diagnosability of $S_{9,3}$, TNR, TPR, ACCR, FNR, FPR, as well as F1 under different faulty nodes.

$ F $	0	20	40	60	80	100	120	140	150
TNR	100%	100%	100%	100%	100%	100%	99.9%	99.9%	99.9%
TPR	100%	100%	100%	100%	100%	97.4%	93.9%	90.2%	84.8%
ACCR	100%	100%	100%	100%	100%	99.4%	98.5%	97.2%	95.5%
FNR	0.00%	0.00%	0.00%	0.00%	0.00%	0.60%	1.80%	3.70%	6.30%
FPR	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.10%	0.10%
F1	100%	100%	100%	100%	100%	98.5%	96.8%	94.8%	91.8%

Table 5

The non-inclusive 1-extra diagnosability of $S_{9,3}$, TNR, TPR, ACCR, FNR, FPR, as well as F1 under different faulty nodes.

$ F $	0	20	40	60	80	100	120	140	150
TNR	100%	100%	100%	100%	100%	99.9%	99.9%	99.9%	99.9%
TPR	100%	100%	100%	100%	100%	99.9%	99.0%	97.8%	95.8%
ACCR	100%	100%	100%	100%	100%	99.4%	98.5%	97.2%	95.5%
FNR	0.00%	0.00%	0.00%	0.00%	0.00%	1.20%	2.90%	5.80%	
FPR	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%	0.00%
F1	100%	100%	100%	100%	100%	99.8%	98.0%	96.1%	92.8%

the theoretical values, TNR, TPR, ACCR and F1 are always close to 1, FNR and FPR approach to 0. As the amount of faulty nodes increases (account for 60), TNR still remains 100%, TPR is higher than 98.5%, ACCR exceeds 99.6% and F1 is more than 99.2%. Meanwhile, FNR is below 0.4%, and FPR is close to 0. With the increasing number of faulty nodes (approximately 120 faulty nodes), as illustrated in Tables 4 and 5, TNR, TPR, ACCR, and F1 exhibit a slight decline, while FNR and FPR demonstrate a relatively upward trend. Notably, TNR remains consistently above 99.9%, and TPR is always higher than 75.4%. Additionally, ACCR continuously exceeds 91.1%, and F1 consistently surpasses 85.9%. Meanwhile, FNR remains below 13.6%, and FPR is nearly 0. These results indicate that our proposed algorithm possesses the good accuracy and self-diagnosis ability for distinguishing faulty and fault-free nodes in a system as the number of faulty nodes increases.

Overall, the experimental results demonstrate a strong alignment with our theoretical analysis, validating the effectiveness of our proposed scheme. Furthermore, the ANFI algorithm introduced in this study exhibits a remarkably high self-diagnosis capability, highlighting its potential for practical applications.

6. Conclusions

The non-inclusive diagnosability as a variant of diagnosability provides more accurate measures for the robustness and reliability of networks than traditional diagnosability. In this paper, we investigate the non-inclusive diagnosability and non-inclusive 1-extra diagnosability of (n, k) -star network $S_{n,k}$ under the MM* model. In addition, the adaptive non-inclusive fault identification algorithm has been designed to conduct experimental analysis, which verifies the correctness of our theoretical results. For a future study, we will explore the non-inclusive diagnosability of more interconnection networks and data center networks.

CRedit authorship contribution statement

Qigong Chen: Writing – original draft, Methodology, Conceptualization. **Jiafei Liu:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Methodology, Funding acquisition, Conceptualization. **Chia-Wei Lee:** Visualization, Supervision, Formal analysis. **Jingli Wu:** Validation, Supervision, Software. **Gaoshi Li:** Supervision, Software, Data curation.

Declaration of competing interest

The authors declare that there is no conflict of interest regarding the publication of this paper. Neither the entire paper nor any part of its content has been published or has been accepted elsewhere. It is not being submitted to any other journal.

Acknowledgement

This work was partly supported by the National Natural Science Foundation of China (No. 62302107 and 62366007), Natural Science Foundation of Guangxi Province (No. 2024JJ170073), Basic Ability Enhancement Program for Young and Middle Aged Teachers of Guangxi, China (No. 2023KY0063), the Science and Technology Project of Guangxi (No. GuikeAD21220114) and National Science and Technology Council of Taiwan (No. 113-2221-E-845-006).

Data availability

No data was used for the research described in the article.

References

- [1] W. Chiang, R. Chen, The (n, k) -star graph: a generalized star graph, *Inf. Process. Lett.* 56 (5) (1995) 259–264.
- [2] E. Cheng, K. Qiu, Z. Shen, On diagnosability of interconnection networks, *Int. J. Unconv. Comput.* 13 (2018) 245–251.
- [3] E. Cheng, K. Qiu, Z. Shen, Diagnosability of interconnection networks: past, present and future, *Int. J. Parallel Emerg. Distrib. Syst.* 35 (2020) 2–8.
- [4] E. Cheng, K. Qiu, Z. Shen, On the g -extra diagnosability of enhanced hypercubes, *Theor. Comput. Sci.* 921 (2022) 6–19.
- [5] T. Ding, M. Xu, Q. Zhu, The non-inclusive diagnosability of hypercubes under the MM* model, *Int. J. Found. Comput. Sci.* 31 (2020) 929–940.
- [6] T. Ding, M. Xu, Relationship between diagnosability and non-inclusive diagnosability of triangle-free connected graphs under the PMC model, *Theor. Comput. Sci.* 944 (2023) 113676.
- [7] S. Dibaker, S. Koushik, Optimal schedule for all-to-all personalized communication in multiprocessor systems, *ACM Trans. Parallel Comput.* 6 (1) (2019) 5.
- [8] K. Feng, Subnetwork preclusion of (n, k) -star networks, *Int. J. Found. Comput. Sci.* 33 (05) (2022) 439–451.
- [9] K. Kumar, B. Sasidharan, K. Sudeep, On oriented diameter of (n, k) -star graphs, *Discrete Appl. Math.* 354 (2024) 214–228.
- [10] X. Li, S. Zhou, X. Xu, L. Lin, D. Wang, The reliability analysis based on subsystems of (n, k) -star graph, *IEEE Trans. Reliab.* 65 (4) (2016) 1700–1709.
- [11] M. Lv, S. Zhou, X. Sun, G. Lian, J. Liu, Reliability of (n, k) -star network based on g -extra conditional fault, *Theor. Comput. Sci.* 757 (2019) 44–55.
- [12] J. Gao, F. Zheng, F. Qi, et al., Sunway supercomputer architecture towards exascale computing: analysis and practice, *Sci. China Inf. Sci.* 64 (2021) 141101.
- [13] J. Huang, E. Cheng, S. Hsieh, Two algorithms for constructing independent spanning trees in (n, k) -star graphs, *IEEE Access* 8 (2020) 175932–175947.
- [14] M. Ma, M. Xu, T. Ding, X. Li, Q. Zhu, The non-inclusion diagnosability of hypercubes under the PMC model, *J. Oper. Res. Soc. China* 12 (2024) 478–484.
- [15] J. Maeng, M. Malek, A comparison connection assignment for self-diagnosis of multiprocessor systems, *IEEE Trans. Comput.* 30 (1981) 173–175.
- [16] S. Peng, S. Yang, W. Su, X. Zhang, T. Zhang, W. Liu, X. Zhao, A CPU/MIC collaborated parallel framework for GROMACS on Tianhe-2 supercomputer, *IEEE/ACM Trans. Comput. Biol. Bioinform.* 16 (2) (2019) 425–433.
- [17] F. Preparata, G. Metze, R. Chien, On the connection assignment problem of diagnosable systems, *IEEE Trans. Electron. Comput.* 16 (6) (1967) 848–854.
- [18] A. Sengupta, A. Dahbura, On self-diagnosable multiprocessor systems: diagnosis by the comparison approach, *IEEE Trans. Comput.* 41 (11) (1992) 1386–1396.
- [19] Y. Wei, T. Ding, M. Xu, The non-inclusive diagnosability of regular graphs, *J. Oper. Res. Soc. China* 11 (2023) 891–910.
- [20] S. Wang, The diagnosability of Möbius cubes for the g -extra condition, *Theor. Comput. Sci.* 908 (2022) 76–88.
- [21] J. Xu, *Combinatorial Theory in Networks*, Science Press, Beijing, 2013.
- [22] D. Xiang, K. Chakrabarty, H. Fujiwara, Fault-tolerant unicast-based multicast for reliable network-on-chip testing, *ACM Trans. Des. Autom. Electron. Syst.* 23 (6) (2018) 73.
- [23] J. Yuan, Y. Li, A. Liu, H. Qiao, The non-inclusive g -good-neighbor diagnosability of interconnection networks, *Theor. Comput. Sci.* 922 (2022) 179–192.
- [24] S. Zhang, T. Tian, Y. Li, C. Ye, Non-inclusive g -good-neighbour diagnosability of hypercubes, *Int. J. Comput. Math. Comput. Syst. Theory* 9 (2024) 54–63.
- [25] N. Zhuo, S. Zhang, C. Ye, Non-inclusive diagnosability of alternating group graphs, *Parallel Process. Lett.* 33 (2023) 2350007.
- [26] S. Zhou, X. Li, J. Li, D. Wang, Reliability assessment of multiprocessor system based on (n, k) -star network, *IEEE Trans. Reliab.* 66 (4) (2017) 1025–1035.
- [27] S. Zhou, Q. Dong, *Reliability Analysis of Multiprocessor Systems*, Science Press, Beijing, 2019.
- [28] H. Zhang, S. Zhou, X. Liu, Z. Yu, Extra (component) connectivity and diagnosability of bubble sort networks, *Theor. Comput. Sci.* 940 (2024) 180–189.
- [29] S. Zhang, W. Yang, The g -extra conditional diagnosability and sequential t/k -diagnosability of hypercubes, *Int. J. Comput. Math.* 93 (3) (2015) 1020796.
- [30] W. Zheng, S. Zhou, E. Cheng, Q. Zhang, Non-inclusive g -extra diagnosability of interconnection networks under PMC model, *Theor. Comput. Sci.* 1022 (2024) 114887.

Sponsor names

Do not correct this page. Please mark corrections to sponsor names and grant numbers in the main text.

National Natural Science Foundation of China, *country*=China, *grants*=62302107, 62366007

Natural Science Foundation of Guangxi Province, *country*=China, *grants*=2024JJB170073

National Science and Technology Council of Taiwan, *country*=Taiwan, *grants*=113-2221-E-845-006